



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0618 – DESIGN AND ANALYSIS OF ALGORITHM

EXPERIMENT NO: 12

Name: Premarubha N

Reg No: 192525020

Program:

```
def merge_count(arr):  
    if len(arr) <= 1:  
        return arr, 0  
    mid = len(arr) // 2  
    left, count_left = merge_count(arr[:mid])  
    right, count_right = merge_count(arr[mid:])  
    merged = []  
    i = 0  
    j = 0  
    count = count_left + count_right  
    while i < len(left) and j < len(right):  
        if left[i] <= right[j]:  
            merged.append(left[i])  
            i += 1  
        else:  
            merged.append(right[j])  
            count += len(left) - i  
            j += 1  
    merged.extend(left[i:])
```

```
merged.extend(right[j:])
```

```
return merged, count
```

```
arr = [2, 4, 1, 3, 5]
```

```
_, result = merge_count(arr)
```

```
print("Number of inversions:", result)
```

Output:

main.py	Visualize	Run ▶	Output
<pre>7 right, count_right = merge_count(arr[mid:]) 8 9 merged = [] 10 i = 0 11 j = 0 12 count = count_left + count_right 13 14 while i < len(left) and j < len(right): 15 if left[i] <= right[j]: 16 merged.append(left[i]) 17 i += 1 18 else: 19 merged.append(right[j]) 20 count += len(left) - i 21 j += 1 22 23 merged.extend(left[i:]) 24 merged.extend(right[j:]) 25 26 return merged, count 27 28 29 arr = [2, 4, 1, 3, 5] 30 31 _, result = merge_count(arr) 32 33 print("Number of inversions:", result)</pre>			Number of inversions: 3